



Obsidian

AUDITS

Tenor Security Review

Auditors

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Introduction

Obsidian Audits

Obsidian audits is a team of top-ranked security researchers, with a publicly proven track record, specialising in DeFi protocols across EVM chains and Solana.

The team has achieved 10+ top-2 placements in audit competitions, placing 1st in competitions for Wormhole, Pump.fun, Yearn Finance, and many more.

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Audit Overview

Tenor is a fixed-rate lending protocol on Morpho Midnight that lets borrowers and lenders automatically renew their loans as they approach maturity: users set their rate, duration, and gating preferences once, and keepers roll the position into a fresh fixed-rate loan at a matching offer.

Tenor engaged Obsidian Audits to perform a security review of the new `FridayEndOfMonthCadence` cadence contract in the `tenor-contracts` repo. The review was conducted on August 14, 2026.

Scope

Files in scope

Repo	Files in scope
<code>tenor-contracts</code> Commit hash: 067bbd10740529b4041dfdf7669f140c0f26138f	<code>src/ratifiers/policies/FridayEndOfMonthCadence.sol</code> (entire file)

Summary of Findings

No issues were identified during this review.

Review Summary

The three internal primitives were verified for correctness:

- `_civilFromDays` (epoch day to calendar date): checked constant-for-constant against Hinnant's `civil_from_days` algorithm (C++20 `std::chrono`). The only deviation is the removal of a negative-day branch that is unreachable under `uint256`.
- `_fridayOnOrBefore` (epoch day to the nearest Friday at or before it): the modular arithmetic was re-derived from the fact that epoch day 1 (1970-01-02) was a Friday, and confirmed to return Fridays exactly, including when the input is itself a Friday.
- `_daysInMonth` (month length): implements the complete Gregorian leap rule. The century exceptions (2100 not leap, 2400 leap) were confirmed correct.

All three `unchecked` blocks were checked for overflow and underflow across the entire reachable input domain, and none is reachable.

Both call sites in `BaseMigrationRatifier` were reviewed, along with the property they rely on: `cadencePeriodStart` never returns a timestamp greater than its input. For variable-rate sources, this guarantees `cadencePeriodStart(block.timestamp)` yields a valid (non-future) renewal period start for rate-policy pricing. For target maturities, it makes `cadencePeriodStart(targetMaturity) == targetMaturity` a precise on-grid check, constraining renewals to the canonical monthly markets (last Friday of the month at the configured time, to the second).

Severity Classification

Severity	Impact: High	Impact: Medium	Impact: Low
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

Impact

- **High** - leads to a significant material loss of assets in the protocol or significantly harms a group of users.
- **Medium** - leads to a moderate material loss of assets in the protocol or moderately harms a group of users. Alternatively, breaking a core aspect of the protocol's intended functionality
- **Low** - leads to a minor material loss of assets in the protocol or harms a small group of users.

Likelihood

- **High** - attack path is possible with reasonable assumptions that mimic on-chain conditions, and the cost of the attack is relatively low compared to the amount of funds that can be stolen or lost.
- **Medium** - the attack/vulnerability requires minimal or no preconditions, but there is limited or no incentive to exploit it in practice
- **Low** - requires highly unlikely precondition states, or requires a significant attacker capital with little or no incentive.